

IEOR 4101: Probability, Statistics and Simulation

Lecture 2: Foundations of Probability

Foundation of Probability: Why?

- Probability is the study of *randomness*
- Uncertainty $\neq$ randomness, but it is one good and effective way to understand and make sense of uncertainty
- Probability is instrumental in:
 - Understanding statistics
 - Building scientific/predictive models

Foundation of Probability

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- Set of *all* possible outcomes is the *sample space*
- ***Probabilities*** are assignments of the likelihoods on these possible outcomes

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 - A single die roll.

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 - A single die roll. Sample space is {1, 2, ..., 6}. A fair die assigns probability $\frac{1}{6}$ to each outcome.
 - Repeatedly throw a fair die until the first six shows up. Sample space is {1, 2, ...}, where k indicates that the first six shows up on the kth roll. Assign probabilities $\frac{1}{6}$, $(\frac{5}{6}) \times (\frac{1}{6})$, $(\frac{5}{6})^2 \times (\frac{1}{6})$, ... to outcome 1, 2, 3, ... (will see why later..)

 {1, 2, 3, 4, ...}

Foundation of Probability

- An ***event*** is *a set* of outcomes in the sample space
- Examples of events:
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 - 3 consecutive coin tosses: what is the subset for the event “no Tail occurs in the 2nd toss”?

sample space: $\{ HHH, H\tau\tau, \dots \}$

event: $\{ HHH, HH\tau, \tau HH, \tau HT \}$

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- An ***event*** is *a set* of outcomes in the sample space
- Examples of events:
 - The event that “an even number shows in a die roll” is the subset $\{2, 4, 6\}$
 - 3 consecutive coin tosses: what is the subset for the event “no Tail occurs in the 2nd toss”?
- One can assign a probability to any event using rules of probability

Specifying Probabilities

- A sample space contains n outcomes. If I pick one outcome in a fully random manner, the probability that I pick outcome i is

$$P(\text{selecting outcome } i) = \frac{1}{n}$$

- Event A consists of k outcomes. If I pick one outcome in a fully random manner, the probability that event A happens is

$$P(\text{event } A) = \frac{k}{n}$$

Algebra of Events

- Further ex: pick a random card from a deck of 52 cards.
 - What is the sample space?
 - What are the events?
 - What are the probabilities?

sample space: $\{1, 2, \dots, 52\}$

events: $\{1\}, \{2\}, \dots, \{52\}, \{1, 2\}, \{2, 3\},$

$\dots \{1, 2, 5, 10, 15\}, \dots$

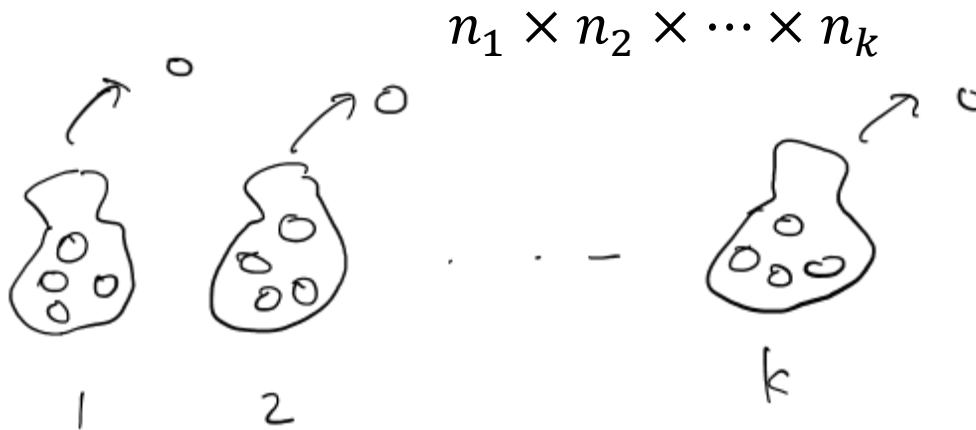
$\{1, 2, \dots, 52\}, \emptyset \leftarrow \text{empty set}$

events: 2^{52}

prob: $P(\{1\}) = \frac{1}{52}, \dots, P(\{52\}) = \frac{1}{52}, P(\{1, 2\}) = \frac{2}{52},$

Counting

- To understand further how to specify probabilities, we also need to know how to count.
- There are k containers. If container i has n_i objects, and all objects are distinct. I pick one object from each container. The total number of possible outcomes is



Counting

- A permutation of n distinct objects is an ordering of these objects
- A permutation of k objects out of n distinct objects is an ordering of any k objects out of the n ones
- The total number of permutations of n distinct objects is
$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 1$$
- The total number of permutations of k distinct objects out of n ones is

$${}_nP_k = n \times (n - 1) \times (n - 2) \times \cdots \times (n - k + 1) = \frac{n!}{(n - k)!}$$

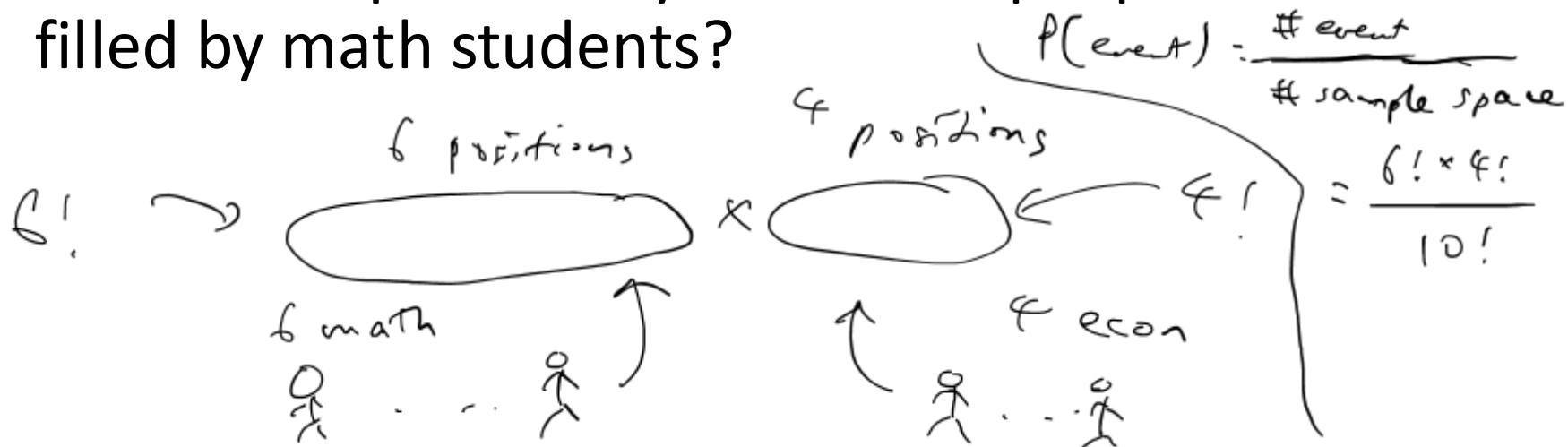
Counting

- A combination of k objects out of n distinct objects is a collection of any k objects out of the n ones (ordering does not matter)
- The total number of combinations of k distinct objects out of n ones is

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

Example

- 4 econ and 6 math students are ranked according to their class performances. Assuming no ties, how many possible rankings are there? $10!$
- Among these rankings, how many have all math students in the top 6 places? $6! \times 4!$
- What's the probability that the top 6 places are filled by math students?



Example

- 4 econ and 6 math students are ranked according to their class performances. Assuming no ties, how many possible rankings are there?
- Among these rankings, how many have all math students in the top 6 places?
- What's the probability that the top 6 places are filled by math students?
- Ans: $10!$, $6! \times 4!$, and $6! \times 4! / 10!$

Example

- A committee of size 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

$$P(\text{event}) = \frac{\# \text{ event}}{\# \text{ sample space}} = \frac{\binom{6}{3} \times \binom{9}{2}}{\binom{15}{5}}$$

$$\# \text{ event} = \binom{6}{3} \times \binom{9}{2}$$

$$\# \text{ sample space} = \binom{15}{5}$$

Example

- A committee of size 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?
- Ans: $\binom{6}{3} \times \binom{9}{2} / \binom{15}{5}$

Example

- A class project requires pairing up two students, among 4 econ and 6 math students. If the pairings are done at random, what is the probability that none of the econ students will work with a math student?

$$P(\text{no econ match with math}) \\ = \frac{\# \text{ event}}{\# \text{ sample space}}$$

sample space = # possibilities of forming S pairs

$$= \frac{\binom{10}{2} \times \binom{8}{2} \times \binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{5!}$$

events = # possibilities where econ group with econ,
math group with math

$$= \frac{\binom{4}{2} \times \binom{2}{2}}{2!} \times \frac{\binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{3!}$$

$$\text{Prob} = \frac{\binom{4}{2} \times \binom{2}{2}}{2!} \times \frac{\binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{3!}$$

$$\frac{\binom{10}{2} \times \binom{8}{2} \times \binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{5!}$$

Example

- A class project requires pairing up two students, among 4 econ and 6 math students. If the pairings are done at random, what is the probability that none of the econ students will work with a math student?

- Ans:
$$\frac{\frac{\binom{4}{2} \times \binom{2}{2}}{2!} \times \frac{\binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{3!}}{\frac{\binom{10}{2} \times \binom{8}{2} \times \binom{6}{2} \times \binom{4}{2} \times \binom{2}{2}}{5!}}$$

Algebra of Events

- We often want to form “composite” events from simpler events
- Example: pick a random card from a deck of 52 cards.
 - A = event that the chosen card is an ace
 - B = event that the chosen card is a heart
 - May be interested in the event that the card is an ace or a heart – $A \cup B$ (read “ A union B ”)
 - Or the event that the card is *both* an ace and a heart – $A \cap B$ (read “ A intersection B ”)

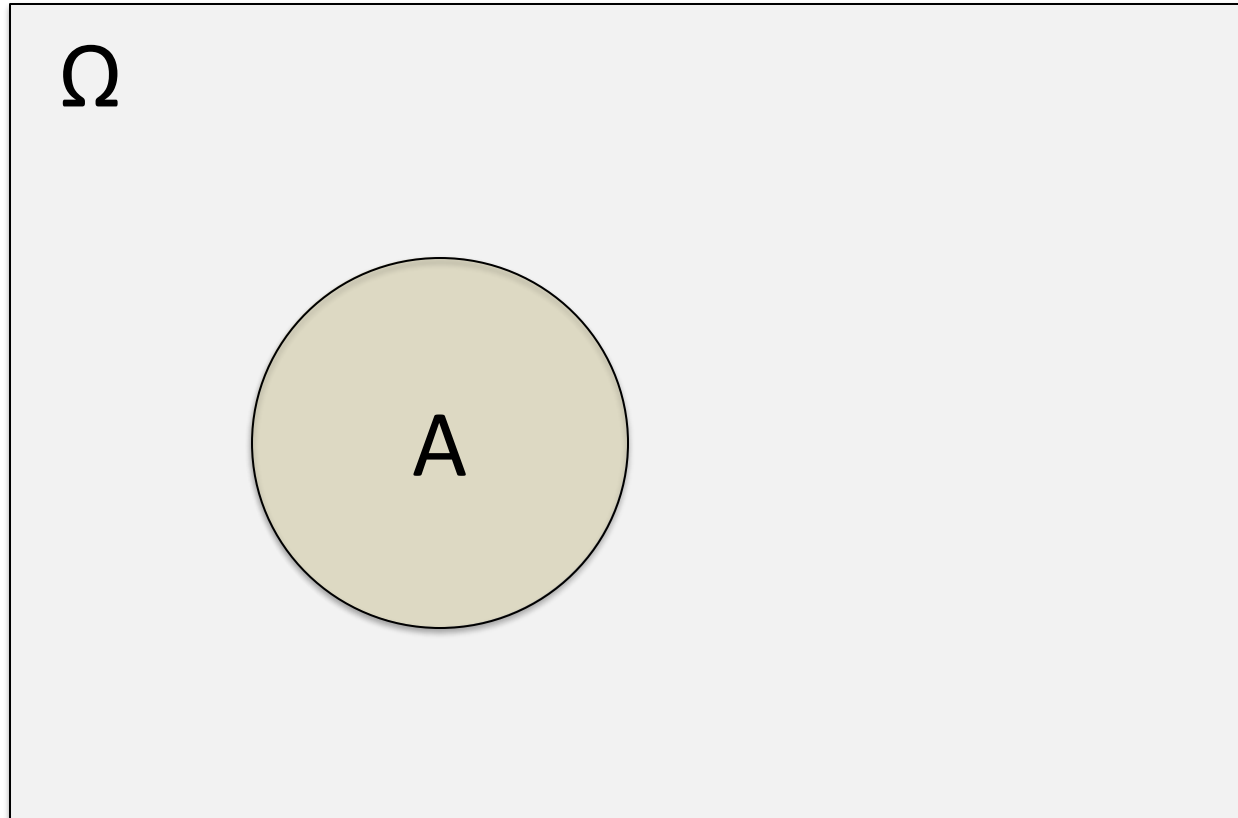
Algebra of Events

- $A \cup B$ is the event that A or B occurs

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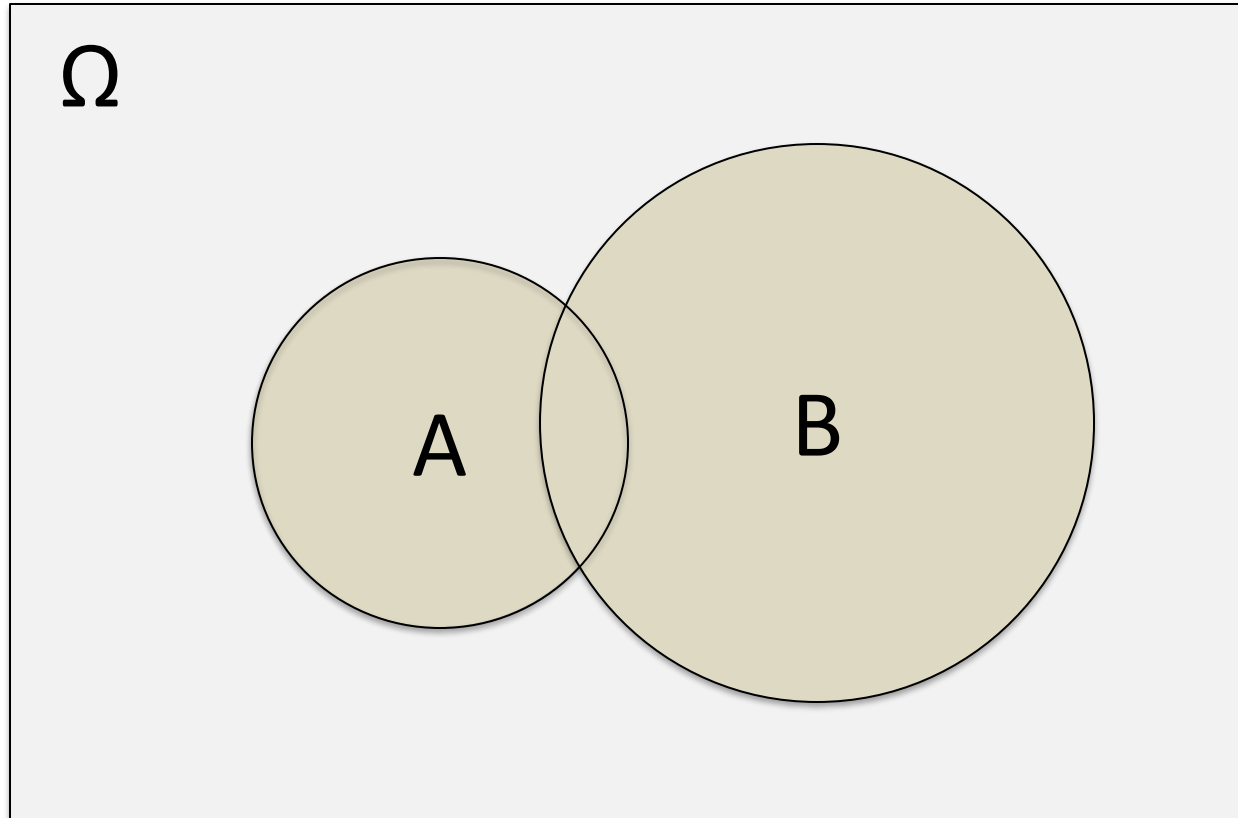
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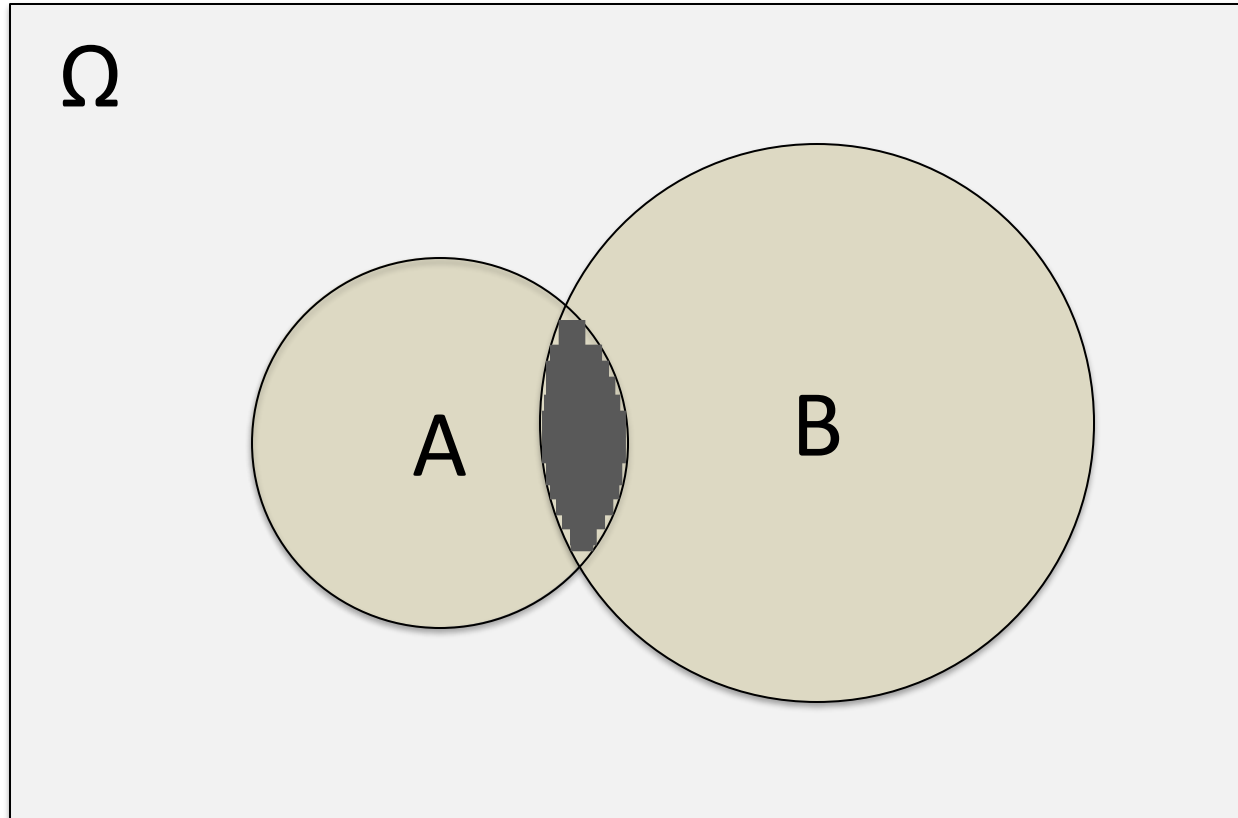
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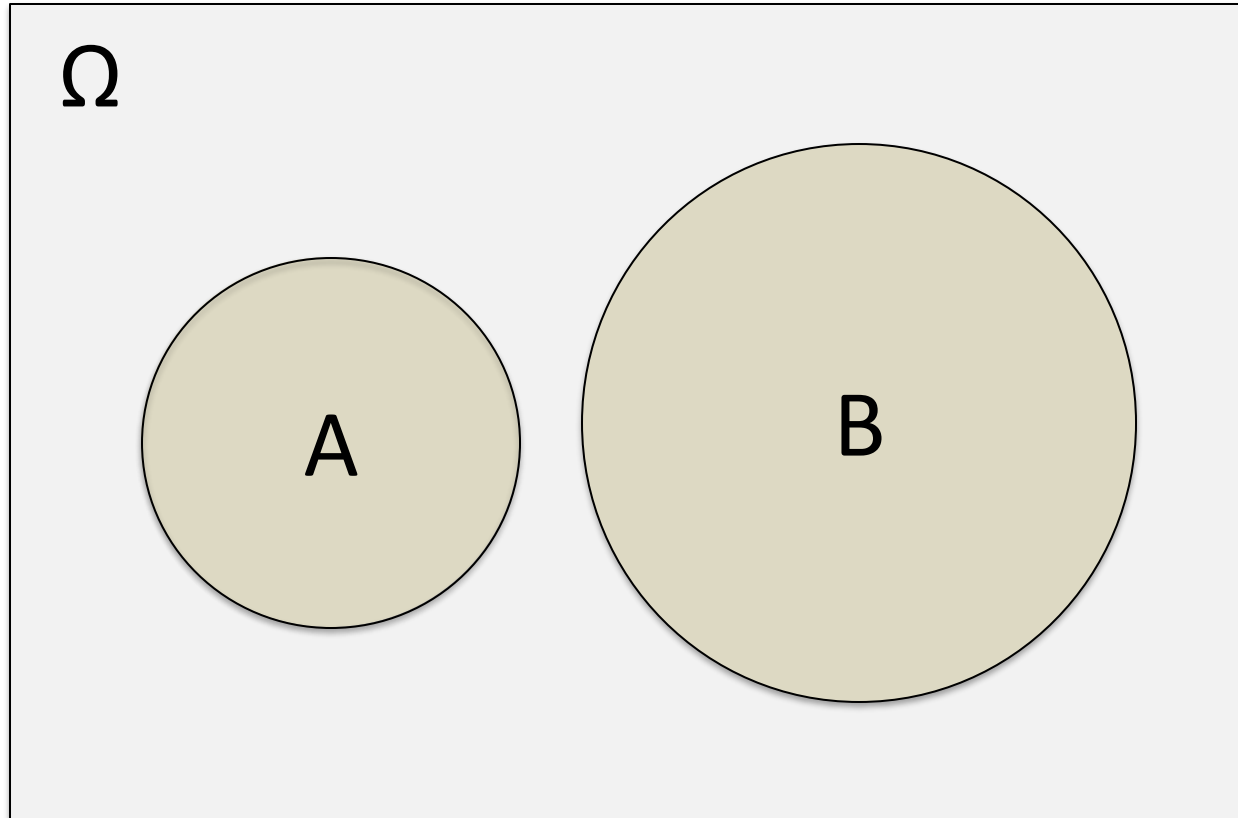
Algebra of Events

- $A \cap B$ is the event that both A and B occur



Algebra of Events

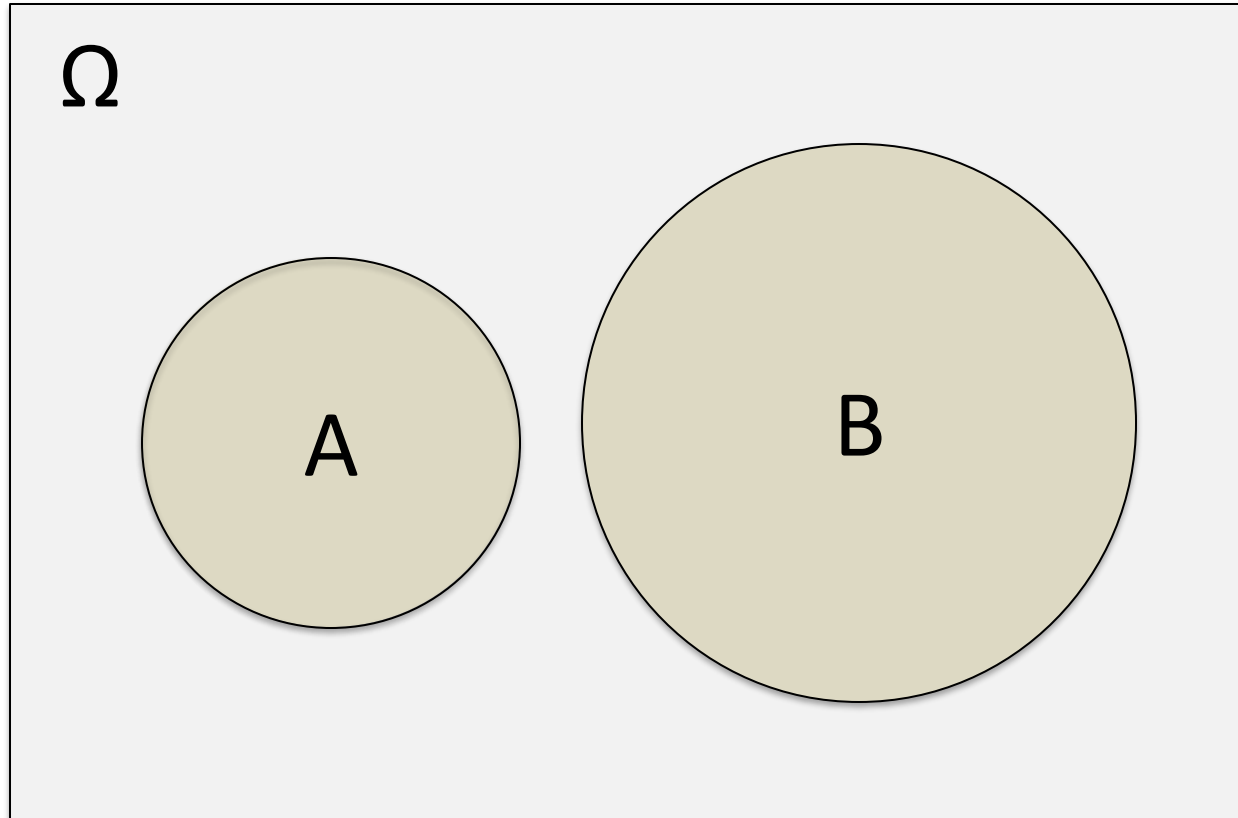
- $A \cap B$ can be *empty*, write $A \cap B = \emptyset$ *empty set*



- In this case, A and B are called disjoint or mutually exclusive

Algebra of Events

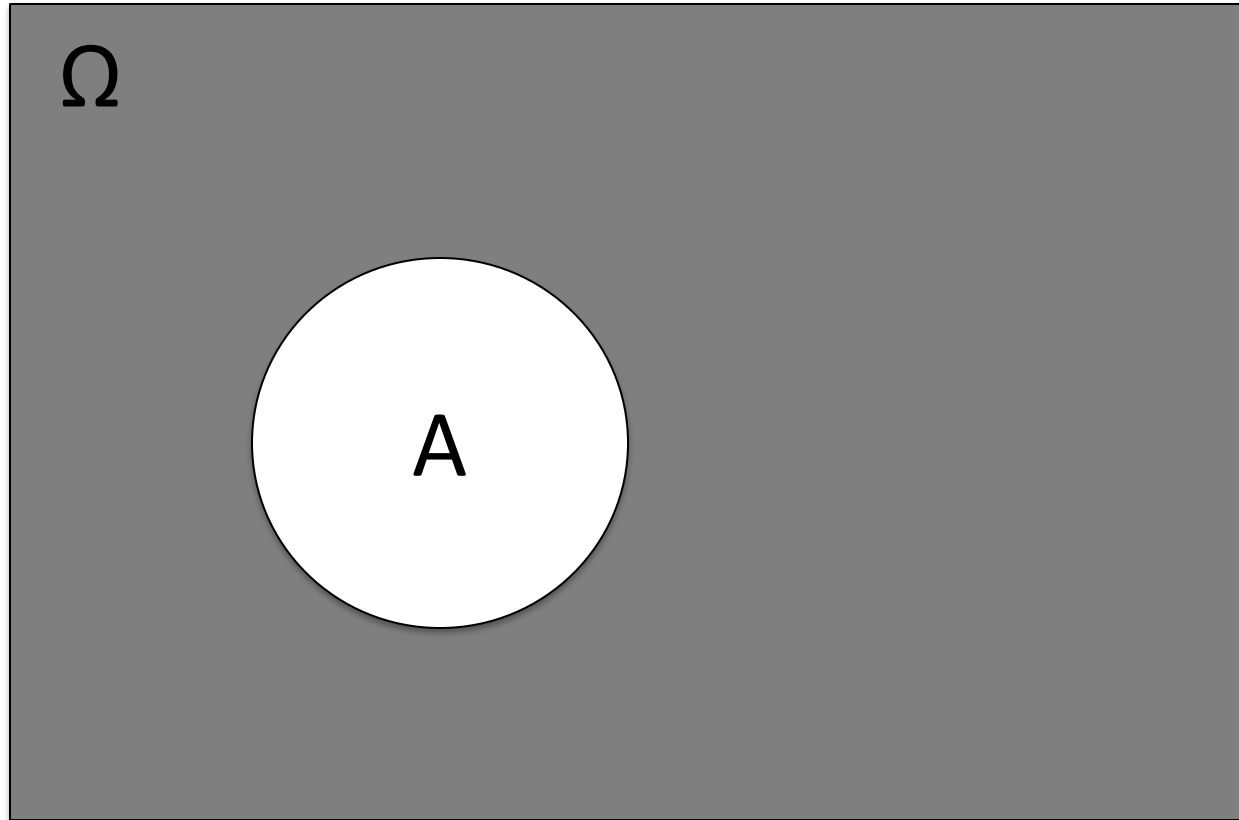
- $A \cap B$ can be *empty*, write $A \cap B = \emptyset$



- In this case, A and B are called disjoint or mutually exclusive
- Q: is the set $\{0\}$ empty?

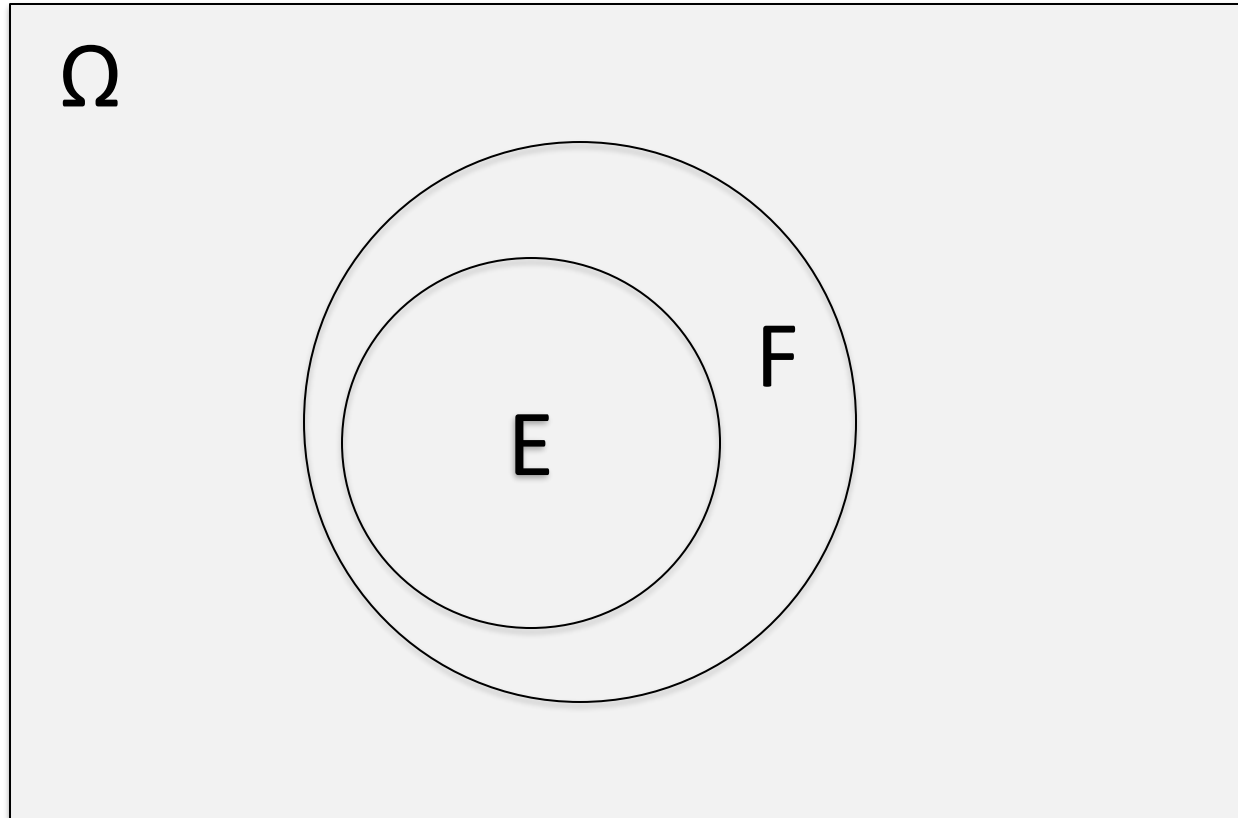
Algebra of Events

- A^c is the complement of A – shaded area



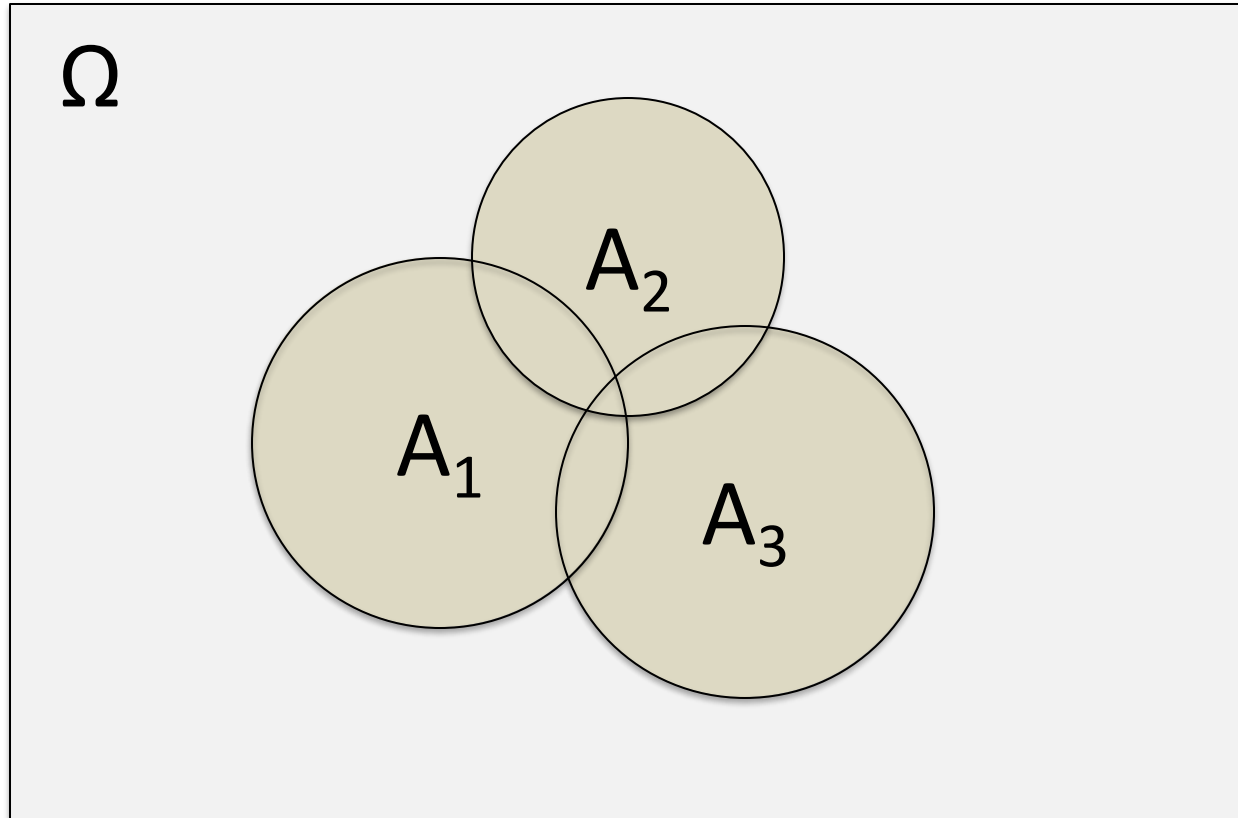
Algebra of Events

- $E \subseteq F$: E is a *subset* of F



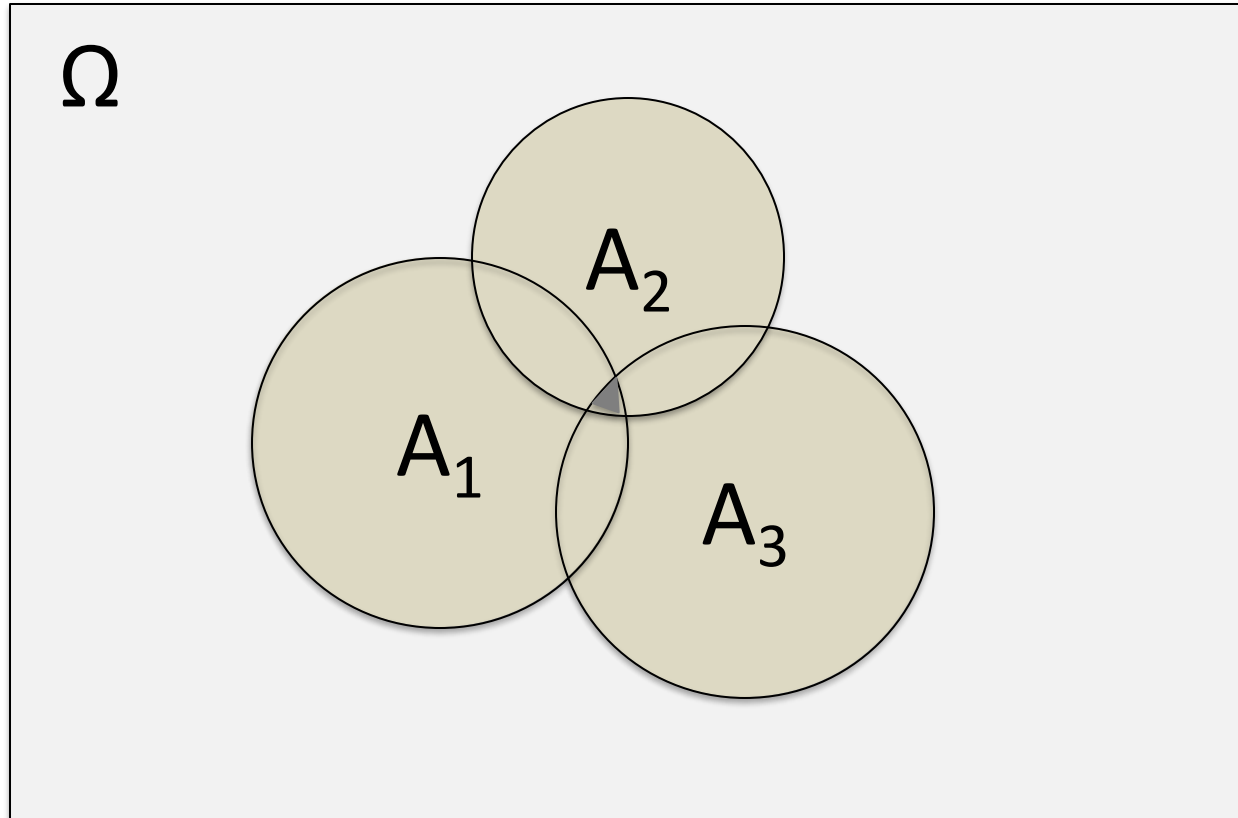
Algebra of Events

- Can also form $A_1 \cup A_2 \cup \dots \cup A_n$, the union of more than 2 events



Algebra of Events

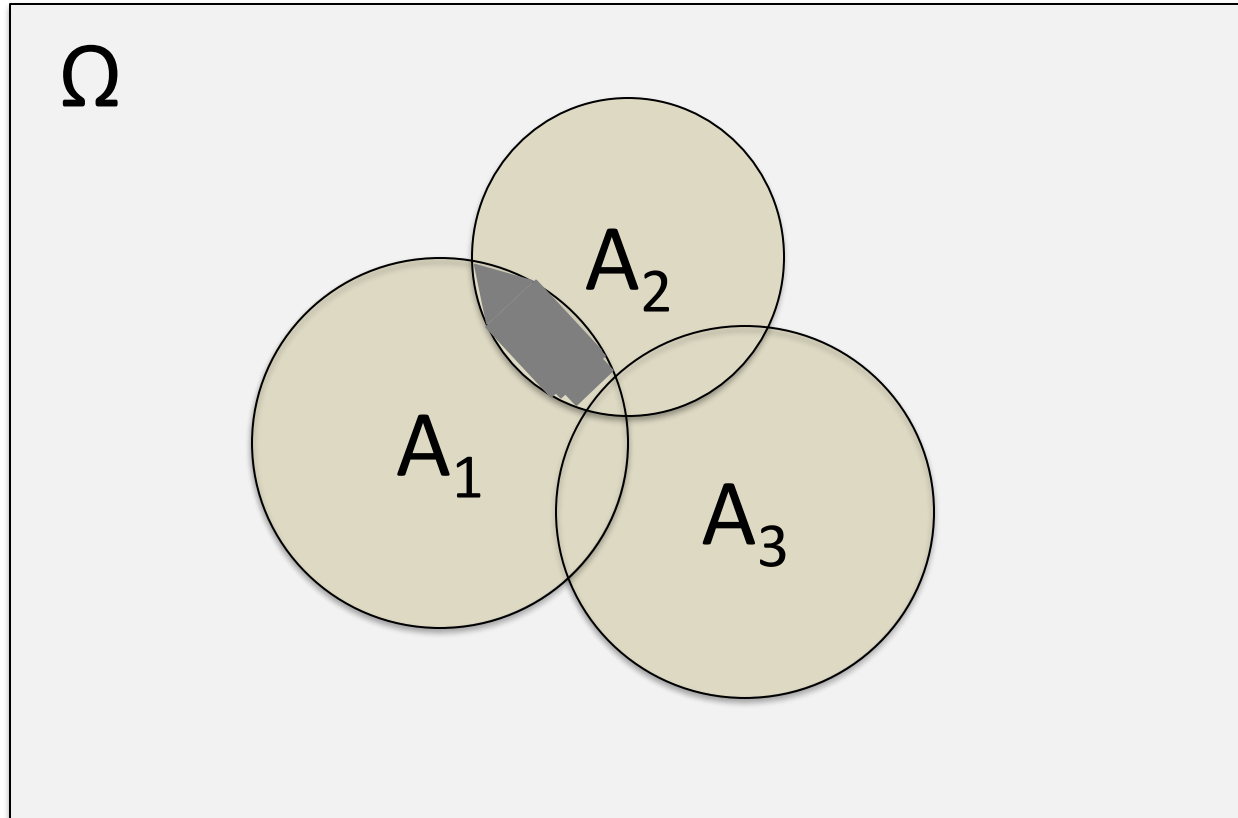
- Can also form $A_1 \cup A_2 \cup \dots \cup A_n$, the union of more than 2 events



- Similarly, can form $A_1 \cap A_2 \cap \dots \cap A_n$, the intersection of more than 2 events

Algebra of Events

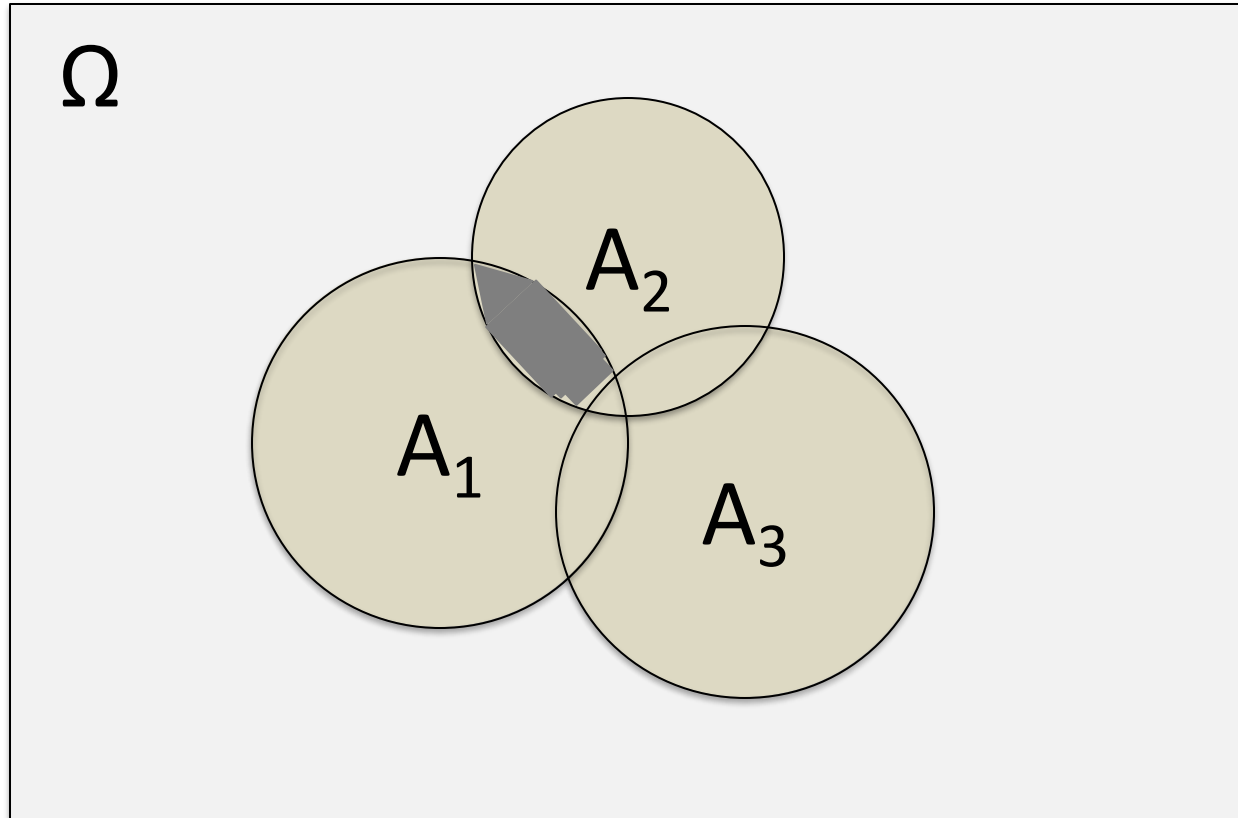
- Q: What is the shaded area in terms of A_1 , A_2 , A_3 , $\cap$, $\cup$, and complement?



$$A_1 \cap A_2 \cap A_3^c$$

Algebra of Events

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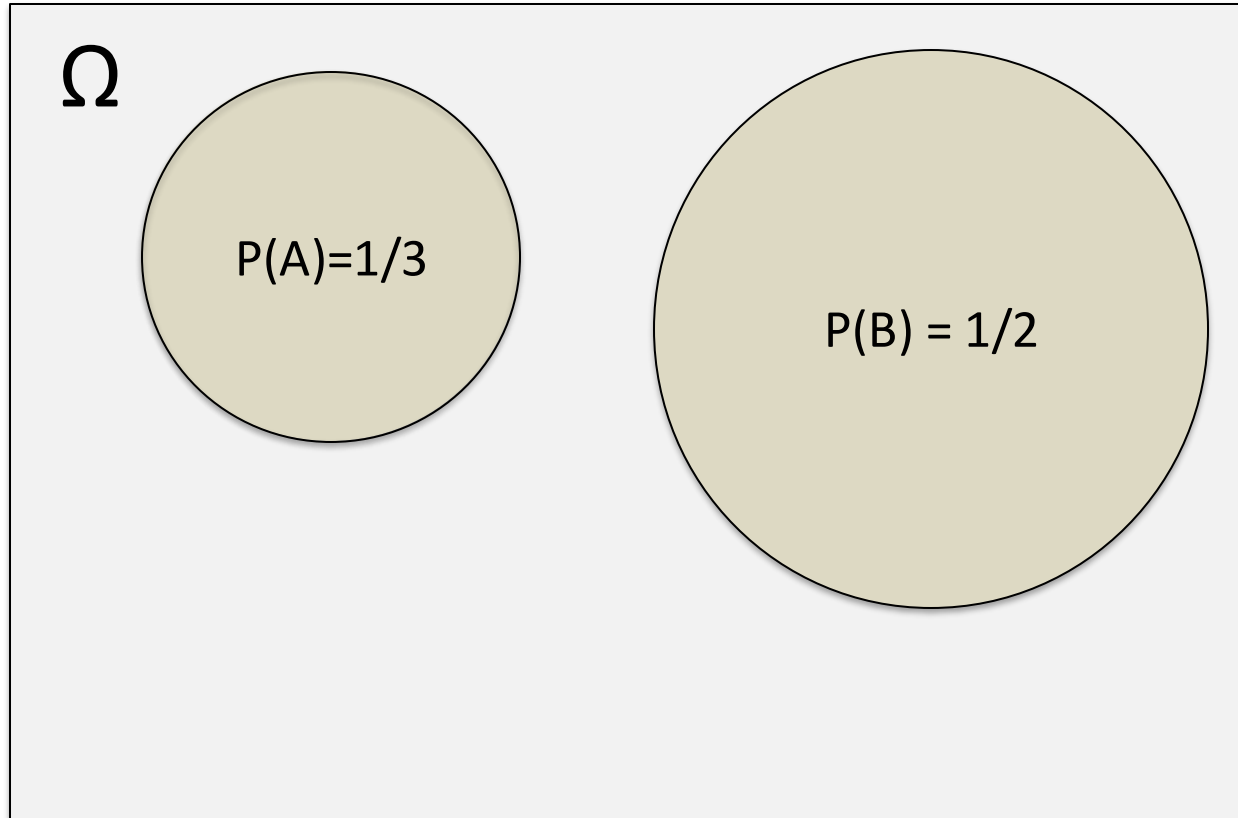
- Ans: $A_1 \cap A_2 \cap A_3^c$

Axioms of Probability

- *Probabilities, events, and sample space* are three pillars of a probability model
- Probabilities are assigned to events of a sample space
- The assignment need to satisfy some properties:
 - **Axiom 1:** $0 \leq P(A) \leq 1$
 - **Axiom 2:** $P(\Omega) = 1$
 - **Axiom 3:** Let $A_1, A_2, \dots$ be mutually exclusive events. Then $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

Axioms of Probability

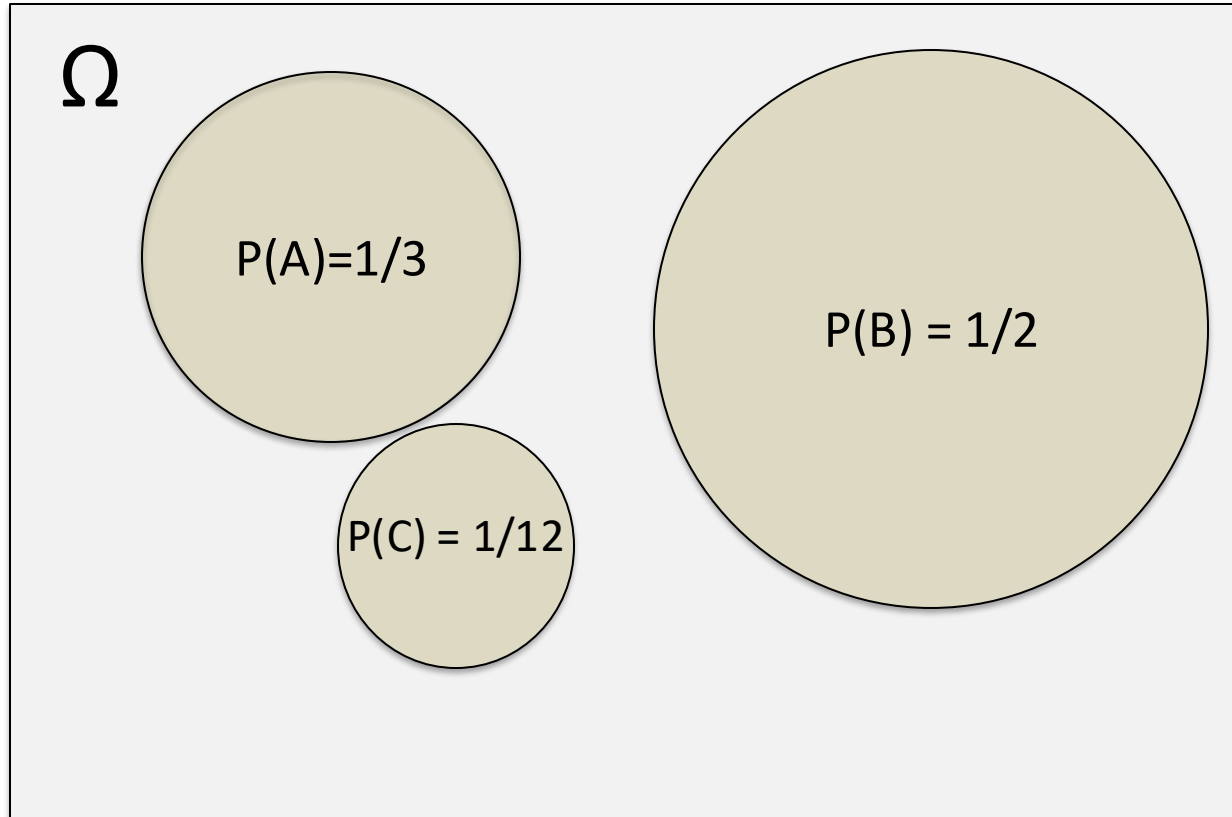
- Illustration of Axiom 3:



- Then $P(A \cup B)$ should be $1/3 + 1/2 = 5/6$

Axioms of Probability

- Illustration of Axiom 3:



- $P(A \cup B \cup C)$ should be $1/3 + 1/2 + 1/12 = 11/12$

Axioms of Probability

- Further properties that we can derive
- **Property 1:** $P(A^c) = 1 - P(A)$
 - Ex. $A =$ “it rains tomorrow,” and $P(A) = 30\%$; then $P(A^c)$, the prob. that it *doesn't* rain tomorrow is $1 - 30\% = 70\%$.
 - *Proof:* A and A^c are mutually exclusive events, so $P(A \cup A^c) = P(A) + P(A^c)$. Also $A \cup A^c = \Omega$, so $P(A \cup A^c) = 1$.
- Often want to compute prob of unions/intersections of events
 - Ex. what is the prob of a randomly drawn card being an ace or a heart?
 - $A =$ event that it's an ace
 - $B =$ event that it's a heart
 - Want to compute $P(A \cup B)$

To show $P(A^c) = 1 - P(A)$,

$$A \cup A^c = \Omega$$

and A & A^c are disjoint

So $P(A \cup A^c) = P(A) + P(A^c)$ by Axiom 3

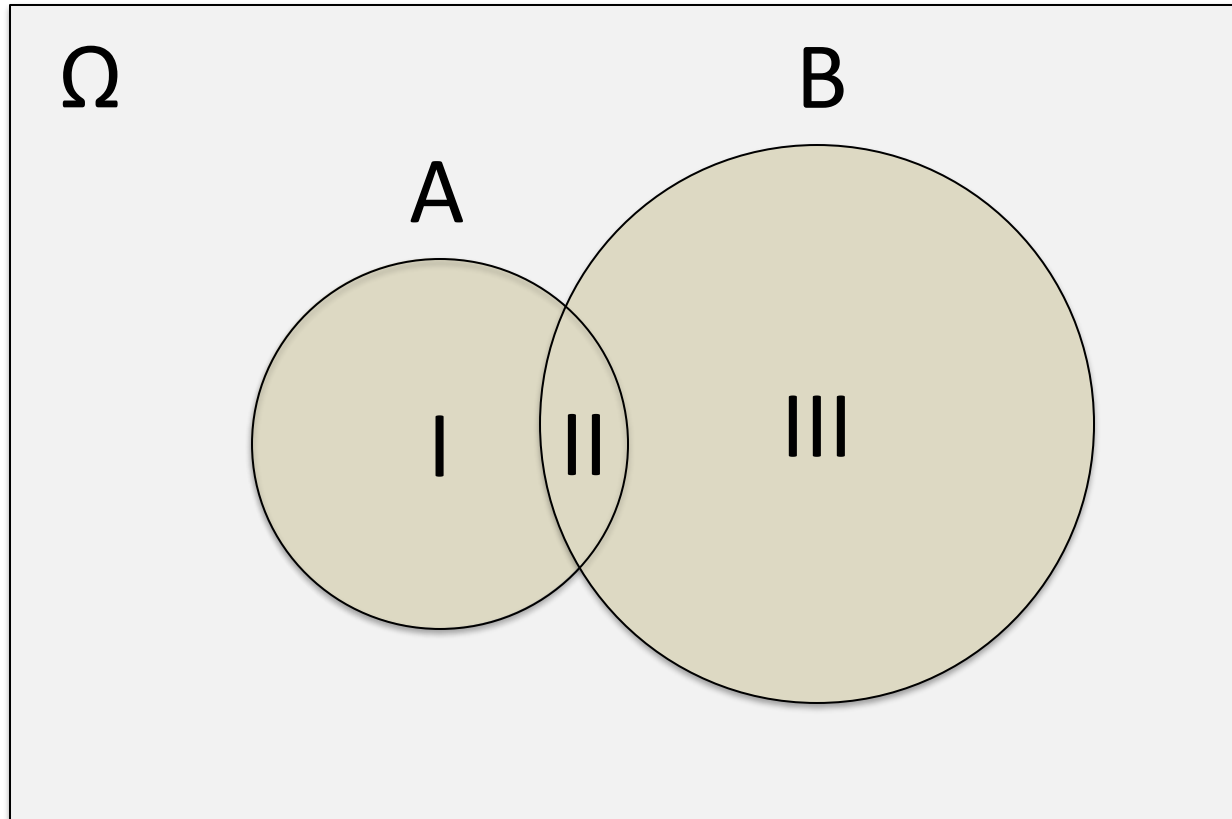
and $P(\Omega) = 1$ by Axiom 2

$$\Rightarrow P(A) + P(A^c) = 1 \Rightarrow P(A^c) = 1 - P(A)$$

Axioms of Probability

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- **Property 1:** $P(A^c) = 1 - P(A)$
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 - *Proof:* A and A^c are mutually exclusive events, so $P(A \cup A^c) = P(A) + P(A^c)$. Also $A \cup A^c = \Omega$, so $P(A \cup A^c) = 1$.
- **Property 2:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 - Pictorial proof on next slide

Axioms of Probability



- $P(A \cup B) = P(I) + P(II) + P(III)$
- $P(A) = P(I) + P(II)$; $P(B) = P(II) + P(III)$
- $P(A \cap B) = P(II)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Axioms of Probability

- Return to example
 - What is the prob of a randomly drawn card being an ace or a heart?

$A = \text{ace}$

$B = \text{heart}$

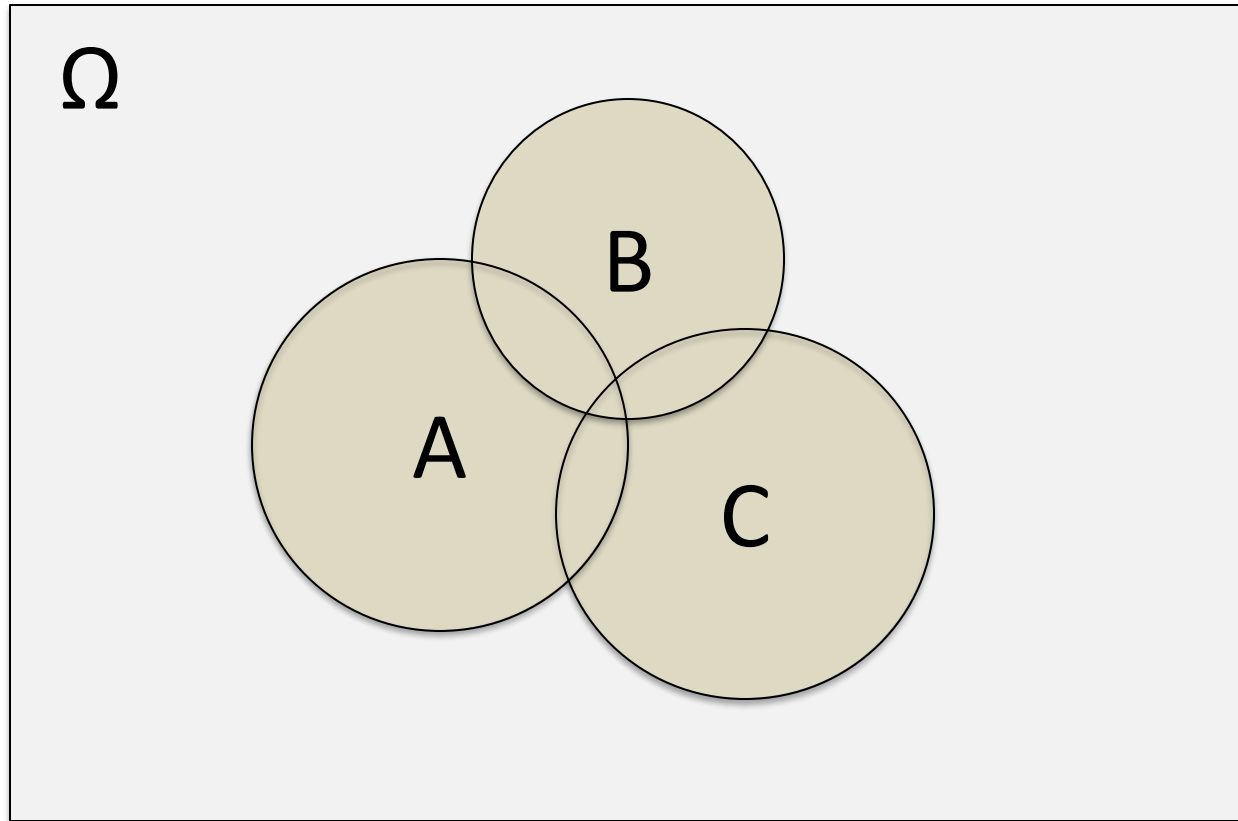
$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} \end{aligned}$$

Axioms of Probability

- Return to example
 - What is the prob of a randomly drawn card being an ace or a heart?
 - A = event that it's an ace
 - B = event that it's a heart
 - $P(A) = 1/13$
 - $P(B) = 1/4$
 - $P(A \cap B) = 1/52$
 - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $= 1/13 + 1/4 - 1/52 = 16/52 = 4/13$

Inclusion-Exclusion Principle

- $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$



Inclusion-Exclusion Principle

- **Q:** The IEOR department has 10 full professors, 2 associate professors, and 7 assistant professors. You will be giving an oral presentation in front of a committee of 4 to be formed randomly. What is the probability that at least one person from each rank is represented on the committee?

$$P(\text{at least 1 from each rank})$$

$$= 1 - P(\text{some rank is not represented})$$

$$= 1 - P(A \cup B \cup C)$$

A = full not represented

B = associate not represented

C = assistant not represented

$$P(A \cup B \cup C) = \dots$$

$$P(A) = \frac{\binom{9}{4}}{\binom{19}{4}}$$

$$P(B) = \frac{\binom{7}{4}}{\binom{19}{4}}$$

$$P(C) = \frac{\binom{12}{4}}{\binom{19}{4}}$$

$$P(A \cap B) = \frac{\binom{7}{4}}{\binom{19}{4}}$$

$$P(B \cap C) = \frac{\binom{10}{4}}{\binom{19}{4}}$$

$$P(A \cap C) = 0$$

$$P(A \cap B \cap C) = 0$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$= \frac{\binom{9}{4}}{\binom{19}{4}} + \frac{\binom{7}{4}}{\binom{19}{4}} + \frac{\binom{12}{4}}{\binom{19}{4}} - \frac{\binom{7}{4}}{\binom{19}{4}} - \frac{\binom{10}{4}}{\binom{19}{4}} - 0 + 0$$

$$A_n = (1 - P(A \cup B \cup C))$$

$$= \dots$$

Inclusion-Exclusion Principle

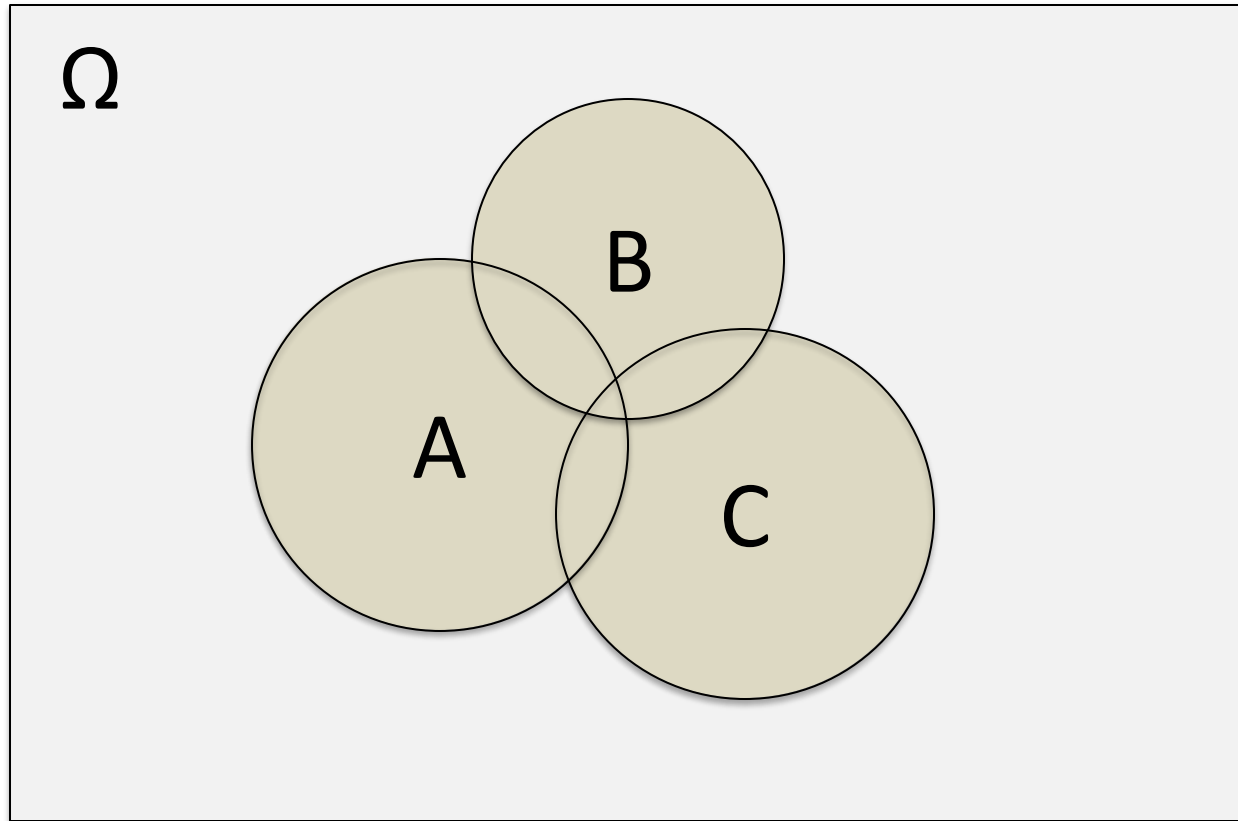
- Label the full professors 1 through 10, associate profs 11 to 12, and assistant professors 13 to 19
- Sample space is the set of combinations $\{p_1, p_2, p_3, p_4\}$, $1 \leq p_1 \neq p_2 \neq p_3 \neq p_4 \leq 19$
- # of outcomes = $\binom{19}{4}$; each outcome is equally likely

Inclusion-Exclusion Principle

- Label the full professors 1 through 10, associate profs 11 to 12, and assistant professors 13 to 19
- Sample space is the set of combinations $\{p_1, p_2, p_3, p_4\}$, $1 \leq p_1 \neq p_2 \neq p_3 \neq p_4 \leq 19$
- # of outcomes = $\binom{19}{4}$; each outcome is equally likely
- A = event that NO full professor is chosen
- B = event that NO associate professor is chosen
- C = event that NO assistant professor is chosen
- $A \cup B \cup C$ = event that at least one category is not represented
- We are interested in $1 - P(A \cup B \cup C)$

Inclusion-Exclusion Principle

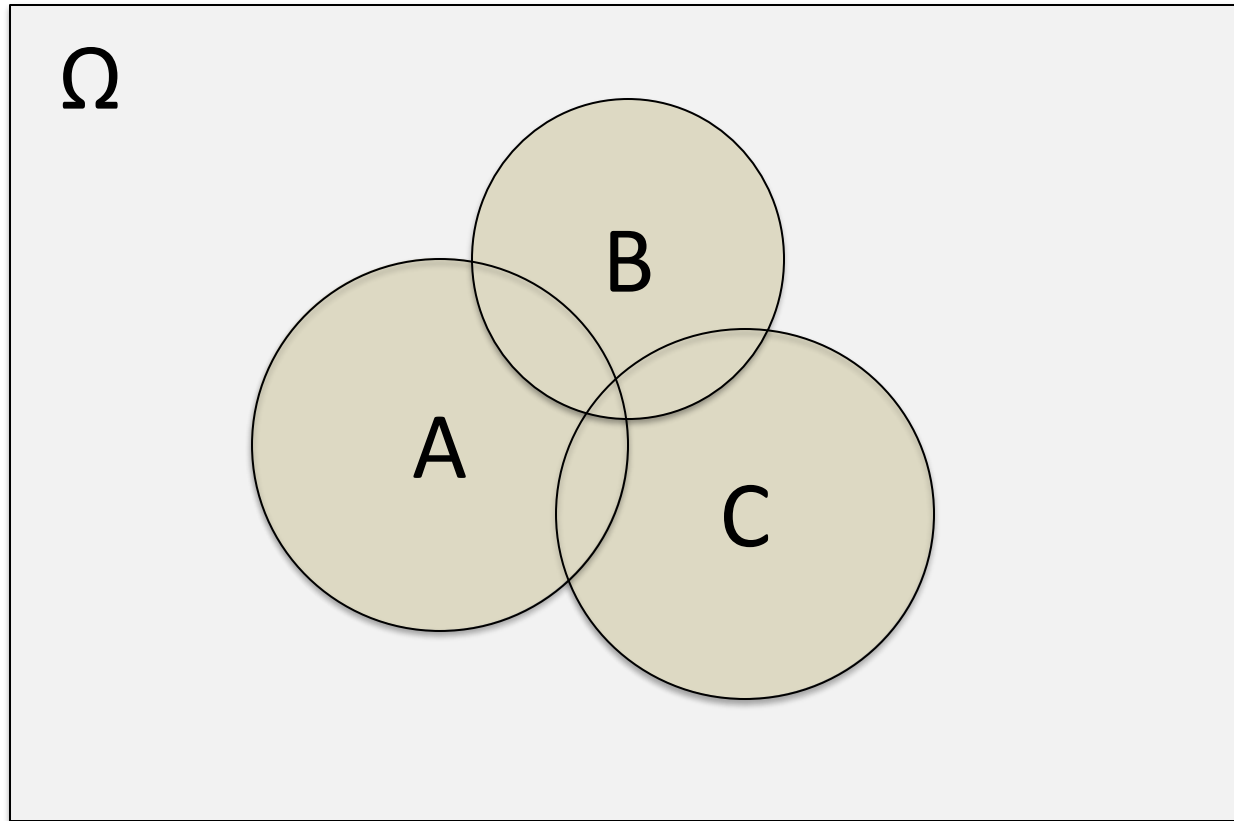
- Can compute $P(A) = \binom{9}{4} / \binom{19}{4}$, $P(B) = \binom{17}{4} / \binom{19}{4}$, $P(C) = \binom{12}{4} / \binom{19}{4}$



- But $P(A \cup B \cup C) \neq P(A) + P(B) + P(C)$

Inclusion-Exclusion Principle

- Can compute $P(A) = \binom{9}{4} / \binom{19}{4}$, $P(B) = \binom{17}{4} / \binom{19}{4}$, $P(C) = \binom{12}{4} / \binom{19}{4}$
- $P(A \cap B) = \binom{7}{4} / \binom{19}{4}$, $P(B \cap C) = \binom{10}{4} / \binom{19}{4}$, $P(A \cap C) = 0$; $P(A \cap B \cap C) = 0$



- $P(A \cup B \cup C) = \dots$